INDIVIDUAL CONTRIBUTION REPORT

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Group number	Group 03
Module	3D Interactive Media
Individual Role:	Student 2 – LEGO Brick Mouse Input Controller System

Introduction

The Lost Island is a 3D adventure game developed using Unity as part of our 3D Interactive Media coursework. Our group consists of four members, and each member was assigned different tasks to contribute towards the development of the game. The main aim of the project is to create an interactive island environment by combining the individual work of all group members.

My assigned role was Student 2, which mainly focused on developing a LEGO brick mouse input controller system. Along with this, I worked on creating a jungle environment and setting up a LEGO minifigure animation system. I developed these features in separate Unity scenes so I could work on them individually and test their functionality before they were added to the main group project.

Through this coursework, I gained practical experience in creating 3D environments, working with imported models, using Unity components and implementing interactive features.

LEGO Brick Mouse Controller

For my main task, I created a scene called Student2_MouseBrick and imported a LEGO brick model. I applied a red material to the brick so that it would be clearly visible in the scene.

I then added Rigidbody and Box Collider components to the LEGO brick and attached a C# script called MouseInput.cs to control its movement. The script was used to allow the brick to respond to mouse input and move when the mouse was dragged in the Game view.

This task helped me understand how mouse input can be used to control objects in Unity and how scripts and components work together to create interactive features.

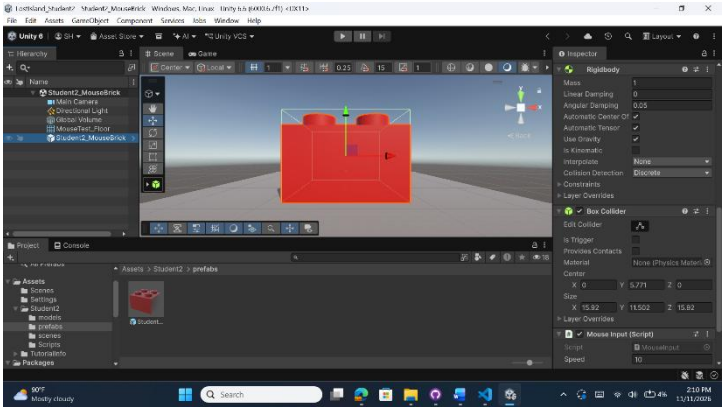


Figure 1:Student 2 Lego brick configured with Rigidbody, Box Collider and MouseInput script.

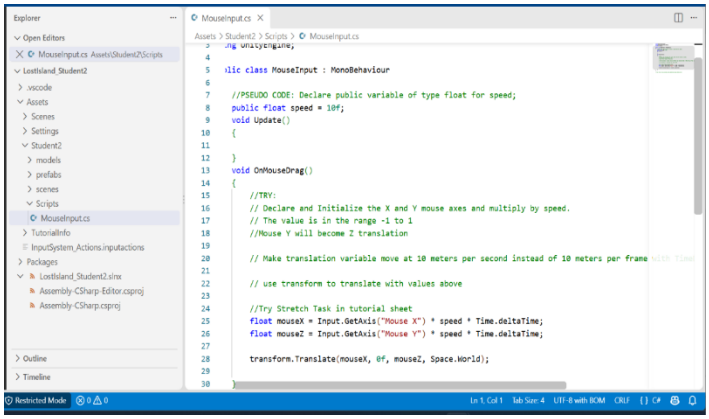


Figure 2:Mouseinput.cs code

Jungle Environment Development

I created a separate scene named Student2_Jungle to represent the jungle exploration area of the island. I started by creating the ground using a Plane and added a pathway using a Cube. I applied green and brown materials to give the environment a natural appearance that matched the island theme.

After setting up the basic environment, I created palm trees using simple 3D shapes such as Cylinders and Cubes. While developing the scene, I made several changes to the placement of objects to make the environment look more organized and realistic. My aim was to create a simple jungle exploration area that would fit into the tropical setting of our group project.

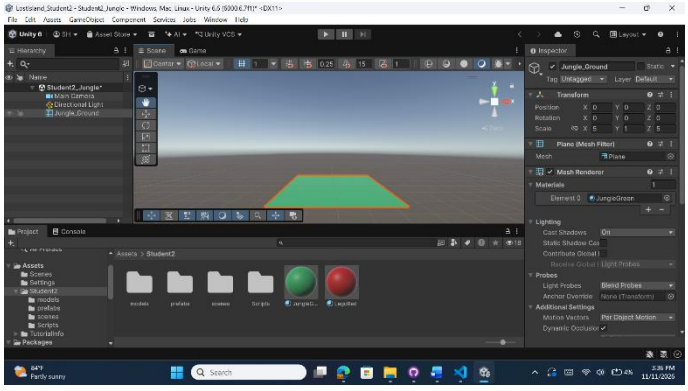


Figure 3: Green Jungle base

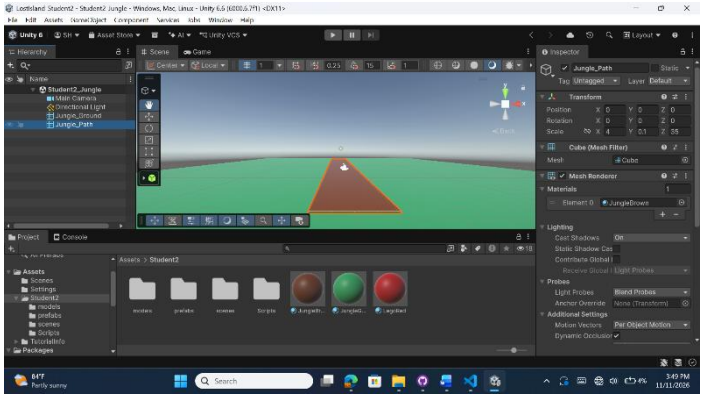


Figure 4:Jungle exploration path for The Lost Island.



Figure 3:First Palm Tree

LEGO Minifigure Animation

Another part of my individual work was creating a LEGO minifigure animation system. I made a separate Unity scene called Student2_LegoAnimation and imported a LEGO character model named LegoWalk.FBX.

Three animations called Stand, Walk and Flip were created using an Animator Controller named LegoAnimator and

added the required animation states.

to this animation I also added three UI buttons labelled Stand, Walk and Flip and connected them to the relevant animations using Unity's Animator Play function. This allowed the user to select an animation by clicking the corresponding button in the Game view.

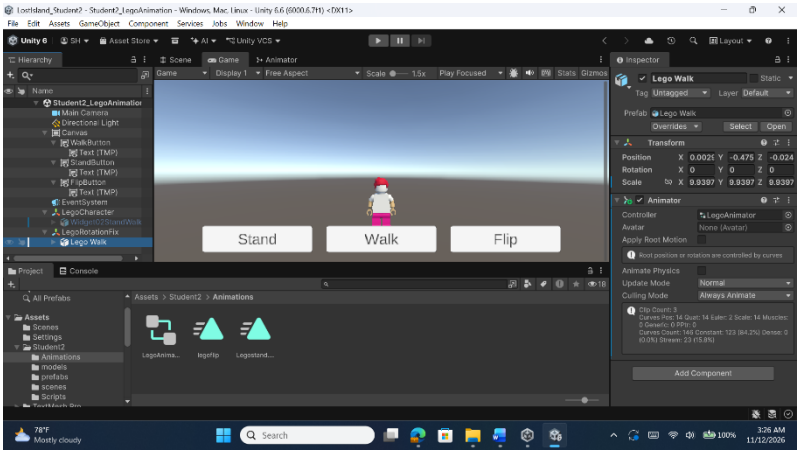


Figure 4:Lego Minifigure with Stand, Walk, Flip.

Testing and Challenges

I tested each feature separately in Unity to make sure it worked as expected. I checked the jungle environment, tested the LEGO brick's movement using mouse input and clicked each animation button to confirm it played the correct animation.

Some challenges included adjusting imported model sizes, positioning objects and fixing animation settings. I resolved these issues by checking the Inspector, changing the necessary settings and testing the features again.

Feature	**Result**
Jungle environment	Created and visible
Mouse input controller	Working
Stand animation	Working
Walk animation	Working
Flip animation	Working

AI Assistance and Personal Reflection

I used ChatGPT during this coursework to understand Unity features and solve problems I faced while developing my tasks. It guided me through creating the jungle environment, setting up the mouse controller and connecting the minifigure animations to UI buttons and errors I faced.

I followed the guidance in Unity and tested the features myself. This coursework helped me improve my understanding of 3D modelling, C# scripts, animation systems and problem-solving. I also became more confident in using Unity and developing interactive features.

References

Unity Technologies. *Unity Documentation*. https://docs.unity3d.com/

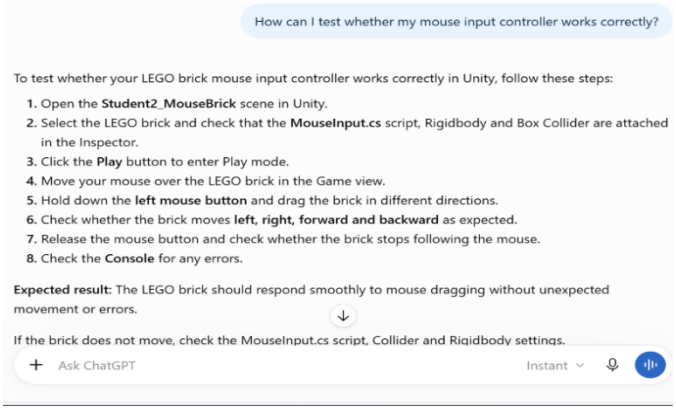
OpenAI (2026). *ChatGPT* https://chatgpt.com/

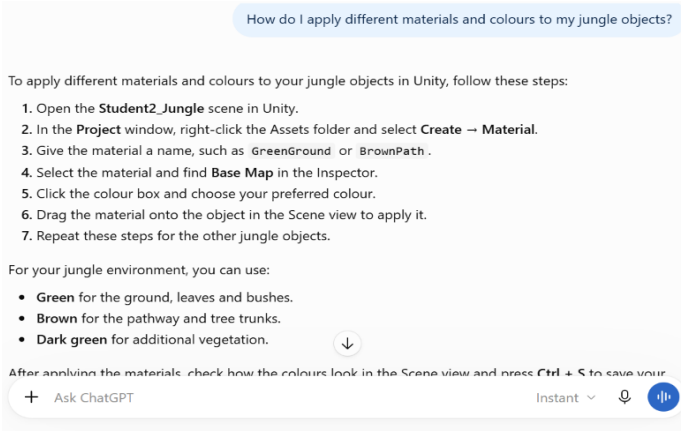
University of Westminster (2026). *3D Interactive Media CW1 Brief and Tutorial Materials (tutorial 02/03) ,(tutorial 5.2)*.

Mecabricks https://www.mecabricks.com/

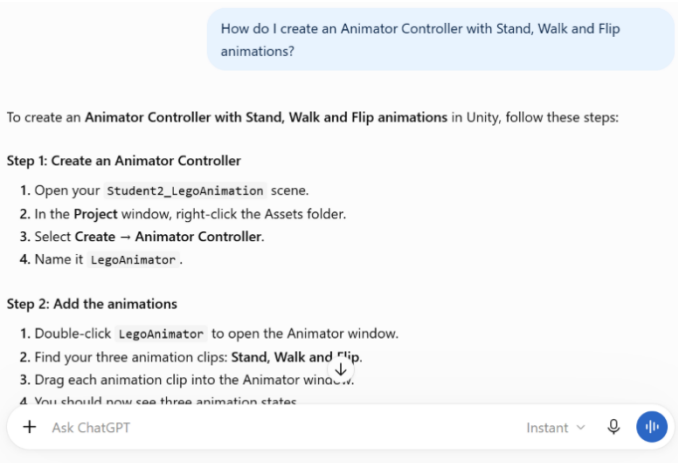
Youtube Animation 101 - Intro to Animation in Unity - YouTube

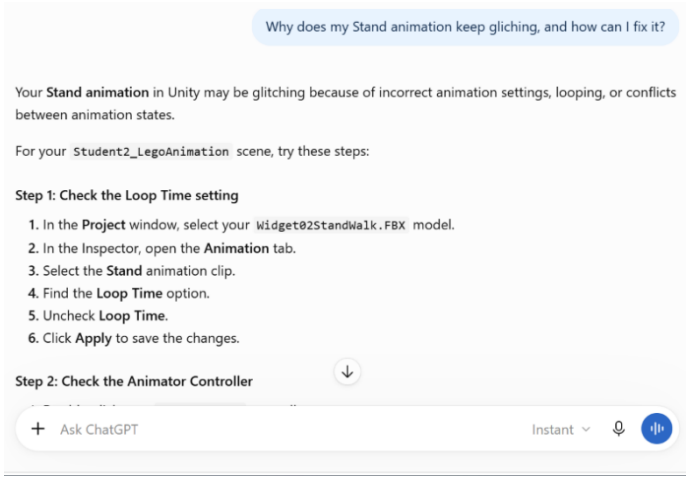
Appendix A





Supporting material – AI transcripts





How can I verify that my Student 2 Unity work meets the CW1 requirements?

To verify that your **Student 2 Unity work meets the CW1 requirements**, check each of your assigned tasks in Unity.

1. Jungle Environment

- Open the `Student2_Jungle` scene.
- Check that the ground, pathway, trees and vegetation are visible.
- Make sure the materials and colours are applied correctly.
- Check that the objects are positioned properly.

2. LEGO Brick Mouse Controller

- Open the `Student2_MouseBrick` scene.
- Check that the LEGO brick has a **Rigidbody**, **Box** **lider** and `MouseInput.cs` script.
- Enter Play mode and drag the brick using the mouse.

Ask ChatGPT Instant